SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA43

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COURSE OUTCOME COVERED

CO-1: Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery. (L2)

Assessment Tool: Case Study

Weightage: 10%

QUESTIONS: **Total Marks: 10**

Scenario: Web Development Standards Implementation

A startup is developing a modern web application. During development, the team encounters multiple issues related to HTML elements, HTTP methods, XML syntax, and web communication protocols.
The following observations are made:

- The team needs to embed a promotional video on the homepage.
- A registration form must securely send user data to the server.
- The navigation menu must follow semantic HTML standards.
- XML configuration files are being used but contain syntax errors.

- The application must communicate over the correct web protocol.

Data Snippets Provided

Snippet 1: Video Embedding Options

A. <media src="video.mp4"></media> B.
<video src="video.mp4" controls></video>
C. <movie src="video.mp4"></movie>
D. <vid src="video.mp4"></vid>

Snippet 2: Form Submission Methods

A. GET
B. POST
C. PUT
D. DELETE

Snippet 3: Navigation Elements

A.
B. <nav>
C. <section>
D. <header>

Snippet 4: XML Syntax

A. <note><to>User</to><from>Admin</from>
B. <note><to>User</to><from>Admin</from></note>
C. <note><to>User</from></note>
D. <note to="User"></note

Snippet 5: Web Communication Protocols

A. FTP
B. HTTPC. SMTP
D. SNMP

Questions:

1. Select the **correct HTML tag for video embedding** and justify your answer.
2. Identify the **appropriate HTTP method for form submission** and explain why.
3. Choose the **proper semantic element for navigation** and justify its usage.
4. Identify the **correct XML syntax** and explain the error in other options.
5. Choose the **correct protocol for web communication** and justify your choice.

Code of Conduct:

I Chunduri Naga Sai Srikanth (192311159) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Chunduri Naga Sai Srikanth

RUBRICS FOR CALCULATION:

Criteria	**Weightage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding ; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts

Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, wellstructured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

Answers

Scenario: Web Development Standards Implementation

1. Select the correct HTML tag for video embedding and justify your answer.

Correct Answer: Option B – <video src="video.mp4" controls></video>

Explanation:

The HTML5 <video> element is used to embed video files directly into a webpage without requiring external plugins.

- The <video> tag embeds video content.
- The src attribute specifies the video file path.
- The controls attribute displays play, pause, volume, and other playback controls.
- It supports video formats such as MP4 and WebM, depending on browser compatibility.
- It improves user experience by allowing users to play and control videos easily.

Example:

```
<!DOCTYPE html>
<html>
<body>
  <h2>Promotional Video</h2>
  <video src="video.mp4" controls width="500"></video>
</body>
</html>
```

Conclusion: The <video> tag is the correct HTML5 element for embedding promotional videos on a website.

2. Identify the appropriate HTTP method for form submission and explain why.

Correct Answer: Option B – POST

Explanation:

The HTTP POST method is used to send data from a client, such as a browser, to a web server. It is commonly used for registration forms, login forms, and other data submission operations.

Advantages of POST:

1. Sends form data in the HTTP request body.
2. Does not place form values directly in the URL.
3. Supports sending larger amounts of data than typical URL-based submissions.
4. Is suitable for creating new records, such as user registrations.
5. Works with HTTPS to protect data during transmission.

Example:

```
<form action="/register" method="POST">
  <input type="text" name="username" required>
  <input type="email" name="email" required>
  <button type="submit">Register</button>
```

</form>

Conclusion: POST is appropriate for registration forms because it sends data in the request body. However, POST alone does not encrypt data; HTTPS is required for secure transmission.

3. Choose the proper semantic element for navigation and justify its usage.

Correct Answer: Option B – <nav>

Explanation:

The HTML5 <nav> element represents a section of a webpage containing important navigation links. These links help users move between pages or sections of a website.

Advantages of the <nav> element:

1. Provides meaningful structure to HTML documents.
2. Helps users navigate the website easily.
3. Improves accessibility for screen-reader users.
4. Makes code easier to understand and maintain.
5. Helps search engines understand the webpage structure.

Example:

```
<!DOCTYPE html>
<html>
<body>
  <header>
    <h1>Company Website</h1>
  </header>

  <nav>
    <a href="index.html">Home</a>
    <a href="about.html">About</a>
    <a href="services.html">Services</a>
    <a href="contact.html">Contact</a>
```

```
    </nav>


</body>

</html>
```

Conclusion: The <nav> element is preferred over a generic because it clearly identifies the navigation section and improves accessibility.

4. Identify the correct XML syntax and explain the error in other options.

Correct Answer: Option B

```
<note>
    <to>User</to>
    <from>Admin</from>
</note>
```

Explanation:

XML (Extensible Markup Language) is used to store and exchange structured data. XML follows strict syntax rules, including proper opening and closing tags and correct nesting of elements.

Explanation of each option:

Option A:

```
<note><to>User</to><from>Admin</from>
```

Error: The <note> element does not have a closing </note> tag.

Option B:

```
<note><to>User</to><from>Admin</from></note>
```

Correct: All elements have matching opening and closing tags, and the elements are properly nested.

Option C:

```
<note><to>User</from></note>
```

Error: The opening <to> tag is incorrectly closed using </from>. XML requires matching tag names.

Option D:

<note to="User"></note

Error: The closing tag is incomplete because it is missing the final > character.

Important XML rules:

1. Every opening tag must have a matching closing tag.
2. XML tags are case-sensitive.
3. Elements must be properly nested.
4. Attribute values must be enclosed in quotation marks.
5. An XML document must contain exactly one root element.

Conclusion: Option B is the only syntactically correct XML document among the given choices.

5. Choose the correct protocol for web communication and justify your choice.

Correct Answer: Option B – HTTP

Explanation:

HTTP stands for Hypertext Transfer Protocol. It is an application-layer protocol used for communication between web browsers and web servers.

When a user opens a website, the browser sends an HTTP request to the server. The server processes the request and returns an HTTP response containing the requested resource or status information.

Working of HTTP:

1. The user enters a website address in the browser.
2. The browser sends an HTTP request to the web server.
3. The server receives and processes the request.
4. The server sends an HTTP response.
5. The browser displays the webpage to the user.

Uses of HTTP:

- Loading web pages and website resources.
- Submitting forms to web servers.
- Communicating between web applications and APIs.

- Transferring text, images, and other web content.

Comparison with other protocols:

Protocol	Main purpose
FTP	Transfers files between systems.
HTTP	Transfers web resources between browsers and servers.
SMTP	Sends email messages.
SNMP	Monitors and manages network devices.

HTTP vs HTTPS:

HTTPS is the secure version of HTTP. It uses TLS encryption to protect data during transmission and helps verify the server's identity.

Conclusion: HTTP is the correct answer for general web communication. For a modern web application handling user registrations and sensitive information, HTTPS should be used to provide secure communication between the browser and server.